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CIVIL AGENCIES COORDINATING ACTIVITIES: GSA-FSS

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DD FORM 1426 (REVERSE)

P.A. ARMY-ER

Other Cust NAVY-YD, AIR FORCE-99

INTERNATIONAL INTEREST

PROCUREMENT SPECIFICATION ASME/ANSI B18.3

SUPERSEDES:

TITLE

SCREW,SHOULDED,HEX-SOCKET HEAD .375,.500,.625 UNC-3A,UNPLATED

MILITARY STANDARD

MS21301

PAGE 1 OF 2

FED SUP CLASS 5305

5305-1734

Ⓐ ENTIRE STANDARD REVISED

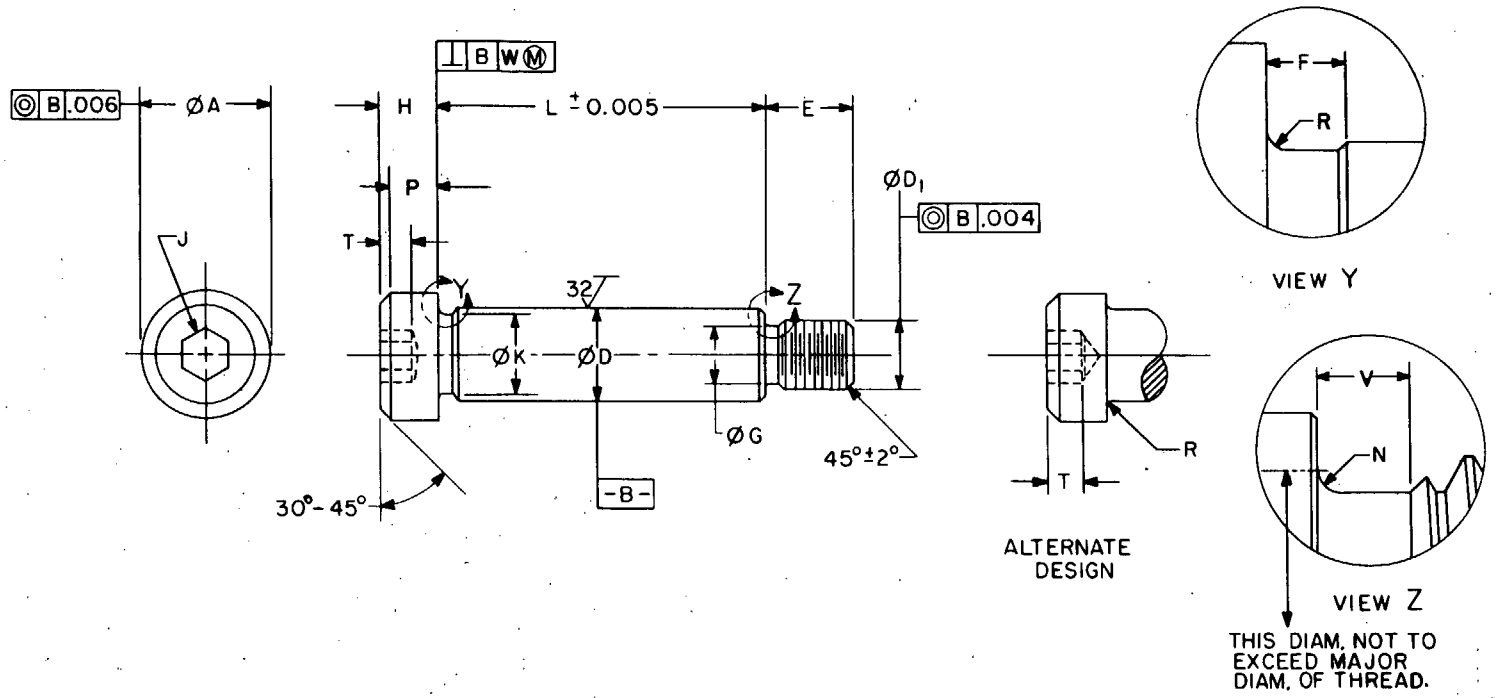


TABLE I. DASH NUMBERS AND DIMENSIONS

NOMINAL SIZE OR BASIC SHOULDER DIAMETER	ØD SHOULDER DIAMETER MAX	ØD SHOULDER DIAMETER MIN	ØA HEAD DIAMETER MAX	ØA HEAD DIAMETER MIN	H HEAD HEIGHT MAX	H HEAD HEIGHT MIN	P HEAD SIDE HEIGHT MIN	J HEXAGON SOCKET SIZE NOM	T KEY ENGAGEMENT MIN	R HEAD FILLET RADIUS MIN	ØK SHOULDER NECK DIAMETER MIN	F SHOULDER NECK WIDTH MAX	ØD1 NOMINAL THREAD SIZE OR BASIC THREAD DIAMETER	ØD1 UNC-3A THREADS/ INCH	ØG THREAD NECK DIAMETER MAX	ØG THREAD NECK DIAMETER MIN	V THREAD NECK WIDTH MAX	N THREAD NECK FILLET MAX	N THREAD NECK FILLET MIN	E THREAD LENGTH MAX	E THREAD LENGTH MIN	W TOLERANCE
.375	.373	.371	.562	.543	.250	.240	.209	.188	.141	.015	.352	.093	.3125	18	.249	.237	.111	.031	.025	.500	.480	.003
.500	.498	.496	.750	.729	.312	.302	.262	.250	.188	.020	.477	.093	.3750	16	.304	.291	.125	.035	.029	.625	.605	.004
.625	.623	.621	.875	.853	.375	.365	.315	.312	.234	.024	.602	.093	.5000	13	.414	.397	.154	.042	.036	.750	.720	.004

APPROVED 18 JUL 58 REVISED Ⓐ 21 DEC 87

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TABLE II. DASH NUMBERS AND DIMENSIONS

NOMINAL SIZE SHOULDER DIAMETER		.375	.500	.625
THREAD SIZE		3125-18 UNC-3A	.375-26 UNC-3A	.500-13 UNC-3A
L SHOULDER LENGTH	TOLERANCE	DASH NUMBER	DASH NUMBER	DASH NUMBER
1.00	±.005	-1		
1.25		-3	-27	
1.50		-5	-29	-59
1.75		-7	-31	-61
2.00		-9	-33	-63
2.25		-11	-35	-65
2.50		-13	-37	-67
2.75		-15	-39	-69
3.00		-17	-41	-71
3.25		-19	-43	-73
3.50		-21	-45	-75
3.75		-23	-47	-77
4.00		-25	-49	-79
4.25			-51	-81
4.50			-53	-83
4.75			-55	-85
5.00			-57	-87
5.50				-89
6.00				-91

REQUIREMENTS

1. MATERIAL: MATERIAL SHALL BE ALLOY STEEL GRADE 4340 (UNS G43400) IN ACCORDANCE WITH AMS 6415 OR MIL-S-5000.
2. PROTECTIVE COATING: UNCOATED (NONE).
3. THREADS: THREADS SHALL BE IN ACCORDANCE WITH FED-STD-H28/2. ACCEPTABILITY OF SCREW THREADS SHALL BE IN ACCORDANCE WITH FED-STD-H28/20, SYSTEM 21.
4. TENSILE STRENGTH: ULTIMATE TENSILE STRENGTH SHALL BE 140 KSI, BASED ON THE MINIMUM THREAD NECK AREA.
5. HARDNESS: HARDNESS SHALL BE IN THE RANGE OF 32 TO 43 HRC.
6. PART NUMBER: THE PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER FOLLOWED BY A DASH NUMBER FROM TABLE II.

EXAMPLE: MS21301-5

DASH NUMBER

BASIC MS NUMBER

MS21301-5 INDICATES - SCREW, SHOULDER, HEX-SOCKET HEAD, .375 SHOULDER DIAMETER; .3125-18 UNC-3A THREAD SIZE; 1.50 SHOULDER LENGTH.

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.
3. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DoDISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.
4. FOR CADMIUM PLATED BOLTS SEE MS51975.

APPROVED 18 JUL 58
REVISED (A) FOR CHANGES SEE PAGES 1 AND 2

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MILITARY STANDARD
MS21301

DD FORM 1 MAY 73 672-1 (COORDINATED)
PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE
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